

Capstone

28–36 minutes

The Capstone in Environmental Studies is intended to bring together the key learning objectives that faculty expect their students to have learned during the course of study. The course examines specific issues related to Environmental Studies, performing research along a pressing theme.

The Environmental Studies Program encourages students to analyze a local or regional environmental issue from multiple perspectives, and notably, encourages some element of environmental activism or community engagement. The hope is that students will engage in a research project that will allow them to gain experience, research skills, and communication skills necessary to succeed in a professional or academic field.

Research Projects

[Housing Affordability and Sustainability for Generation Z in Austin \(Fall 2024\)](#)

Project Summary: Many members of Generation Z view homeownership, especially in North America's most desirable cities, as utterly unattainable. Due to this sentiment, this capstone paper explains the connections between economics, policy,

accessibility, justice, and environmental issues surrounding housing for Gen Z. Because of our proximity to Austin and because Austin continues to be one of the most attractive large cities in the U.S. for Gen Z workers, capstone students chose the city of Austin as a case study to help focus their research. The goal of their research is to empower Gen Z with the knowledge to make choices regarding the housing market, as well as provide practical recommendations for different ways members of Gen Z and people of all ages can become involved in their communities to improve access to affordable and sustainable housing for all.

Completed By: A.J. Evanczyk, Michael Purdom, Rose Reed, Sophia Trifilio, Sarah Ventimiglia, Hailey Vickich

[From Blueprint To Greenprint: Sustainable Recommendations For The SU 560 Project \(Fall 2024\)](#)

Project Summary: In 2022, Southwestern University (SU) announced its plan to develop more than 560 acres of land adjacent to its historic campus, a project that will require many years and millions of dollars in investment from the University. This project needs to consider advancements in ecological and social welfare in combination with economic benefits. While recognizing the importance of development, several practices should be incorporated into the 560 Project's plan to promote environmental, economic, and social welfare. This report proposes a set of best practices that capstone students have deemed most beneficial, effective, and feasible to ensure that the 560 Project is accomplished such that it not only mitigates environmental harm but also promotes ecological harmony.

Completed By: Logan Ferguson, Meredith Lannan, Isabella Moore, Sydney Strange, Brianna Westmoreland, Abby White, Blaine ten Wolde

[The ecological, economic, and cultural impacts of Eastern Oyster population decline: a case study of the Texas Coast \(Fall 2023\)](#)

Project Summary: Oysters help maintain stable marine conditions by filtering water, regulating nutrients, protecting surrounding habitats from wave disturbances, and stabilizing the structure of the physical environment. This species supports fish, crabs, invertebrates, other macrofauna, and microfauna. While it has been largely overlooked, the decline in Eastern Oyster numbers is becoming one of the most pressing environmental concerns in Texas' fisheries today. The current population's declining health will impact not only the direct market but also other fundamental coastal industries. To properly understand the issues with the management of the oyster population, this case study examines the environmental, economic, and social aspects of the Texas Coast.

Completed By: Delaney Anthony, Claire Haughey, Oliver Johnson, Lauren Plowman, Emery Stobaugh, and Catherine Yow

[A Peek Beneath the Surface of Texas Groundwater Use \(Fall 2023\)](#)

Project Summary: Texas cities rely on groundwater in an abundance of ways, from human culture to agriculture. Over the past decade, groundwater stores in Central Texas have been on a

notable decline, with iconic spring-fed landmarks such as Jacob's Well in Wimberly consecutively going dry in the summers of 2022 and 2023. Of course, the state of water security in Central Texas is not tied to one source, but has interlocking roots in the political, social, and geologic conditions of its inhabitants. Using the recent partnership between Georgetown and EPCOR as a case study, capstone students examine how the private acquisition and distribution of groundwater in Central Texas has evolved in the twenty-first century, and how our changing relationship to groundwater manifests in legislation, the environment, and our affected communities, both urban and rural.

Completed By: Jackson Legrande, Mace Wages, Max Mazoch, Raquel Kurth

[Urban Ecological Infrastructure for Stormwater Irrigation on a College Campus \(Fall 2022\)](#)

Project Summary: This paper explores the effects of Urban Environmental Infrastructure (UEI), Blue Infrastructure (BI), and Green Infrastructure (GI) on college campuses to maximize water quality and health at universities and within their general watershed regions. Throughout this paper, students outlined how university campuses are a prime example of institutions experiencing growth at levels that negatively impact surrounding watersheds. Students propose recommendations to use these ecologically friendly infrastructures, including bioretention practices, permeable pavements, and swales on university campuses. Southwestern University in Georgetown, Texas, is a primary focus of this paper due to it being an effective candidate

for implementing these infrastructures as well as the development of the currently empty 500-acre plot of land.

Completed By: Ceanna Cooksey, Emily Funk, Anthony Garcia, Gabrielle Garza, Jaime Hotaling, Krista Kelly, Sarah Parks, Katherine Stothoff, Alex Thompson

[Uncovering the Truth about Recycling: Scaling down Recycling's Impact to Southwestern University \(Fall 2022\)](#)

Project Summary: This research paper aims to better understand the state of the current American recycling system. Key attention is paid to 1) accessibility and education in relation to recycling services, 2) behaviors and attitudes regarding the practice of recycling, and 3) the environmental justice and economic implications of the practice of recycling. In combination with the above background research, these capstone students administered a Southwestern University campus-wide survey that aimed to analyze connections between recycling rates and individual demographics of students, faculty, and staff. This paper also connects past and current education about recycling and recycling practices, recycling habits and behaviors, and additional related factors.

Completed By: Megan Crinklaw, Alexis Guajardo, Sydney Kattawar, Abby McGee

[Below the Surface: An examination of health effects of Cyanotoxins and surrounding policy in Central Texas \(Fall 2022\)](#)

Project Summary: Toxin-producing algae blooms known as

cyanobacteria are becoming increasingly prevalent in our waterways. Cyanobacteria can produce toxins that can affect various parts of the body. For example, different species of cyanobacteria can produce toxins that can damage the liver (hepatotoxins), affect the central nervous system (neurotoxins), and produce toxins that affect the renal system and gastrointestinal tract. In this paper, students inquire into the prevalence of cyanobacteria concerning animal and human health. Furthermore, students investigate the lack of cyanobacteria mitigative policy implementation in local waterways.

Completed By: Garrison Engstrom, Dawson Gonzales, Samuel Irizarry, Chloe Penrod, Chase Thinger

[A \(Un\)Democratic Petrostate: Oil Wealth, Corruption, and Exploitation in Texas \(Fall 2021\)](#)

Project Summary: In a country with such high levels of consumption, Texas stands out as a principal source of energy production. Texas is recognized as the top producer of oil and gas within the country. This has led to a complicated and intertwined relationship between oil production, the state economy, and a Texas culture dedicated to the spirit of the frontier. Through manipulating this culture and making oil seem thoroughly more relevant to the national identity and economy than it actually is, the Texas Railroad Commission has continued to exploit Texans and increase their authority in an undemocratic means. This article further examines the ways in which the Texas Railroad Commission has shaped Texas as a petrostate through means that can be described as anti-democratic. It will draw from theories

of political ecology, authoritarianism, petropolitics, necropolitics, and environmental justice in order to argue that the Texas Railroad Commission has played an active role in degrading the air, water, and land quality of Texas in ways that are detrimental to already underrepresented Texans.

[Sustainable Residence Hall Design and Construction: A Proposal for Southwestern University \(Fall 2021\)](#)

Project Summary: Southwestern's Tactical and Master plans propose a new residential hall to be built in the near future, and with these exciting new developments comes an opportunity to renew its position as a leader in sustainable campus living. It is crucial that universities not only aim to promote sustainable campus infrastructure, but also work to produce graduates who understand the importance of integrating sustainability into every aspect of their lives. This report relies upon the diverse literature on green building practices and sustainable design, while also drawing from real world examples and case studies from other institutions of higher education. In this report, students organize their findings in six key themes: (1) Green Building Certifications, (2) Sustainable Building Materials, (3) Sustainable Design and Innovation, (4) Inclusive Housing, (5) Student Engagement and Retention, and (6) Educating Students.

Completed By: Sage Clay, Olivia Gray, Brooke Guthrie, Darcy Hansen, Cameron Mathis, Ricky Olivares, Keith Rupe, Megan Schlab, Nicole Rajtak

[U.S. Agriculture in the Age of COVID-19: From Farm to Fork \(Fall](#)

[2020\)](#)

Project Summary: The American food sector has consistently struggled to meet the nutritional demands of the entire American populace while also protecting the well-being of its employees and the environment. This can be attributed to a number of factors, chief among them income inequality, inefficiency in the supply chain, and worker vulnerability. COVID-19 brought new challenges to the food sector, spurring changes in consumer behavior, public health guidelines, and working conditions, as well as exacerbating already existing problems. Given the need to fortify the food sector against future crises, this paper seeks to explore the effects of the COVID-19 pandemic on the U.S. agricultural supply chain at four key stages: farming, processing, retail, and consumer/public health, or from “farm to fork.” This paper uses a food justice lens, with a focus on waste, labor, and access, to review the many impacts of COVID-19 on the U.S. food supply chain and suggest potential avenues for future research.

Completed By: Nissi Frutos, Ethan Hendrix, Leah Horick, Sophia Potempa,
Abby Prilliman, Sam Ramon, and Emmett Tassin

[#BeeSouthwestern: Bee Campus USA Certification & Pollinator Protection at Southwestern University_\(Fall 2019\)](#)

Project Summary: Bee Campus USA is an initiative of the Xerces Society for Invertebrate Conservation that provides recognition for universities in the U.S. that are committed to preserving local biodiversity through protecting native pollinator habitats, limiting

the use of synthetic pesticides, and engaging in public awareness initiatives. Students found that Southwestern had already met a majority of the requirements for Bee Campus USA. With the absorption of Bee Campus USA duties into the Sustainability Committee, the installation of signage and planting of seeds and starter plants by this group, the creation of a Bee Campus USA webpage on the SU website, the requirements of certification will be met, and the certification will be renewed annually by the Sustainability Committee.

Completed By: Samantha Buehler '20, Karonech Chreng '20, Katey Ewton '20, Spencer Kleypas '20, Abigail Lloyd '20

- [Southwestern Becomes the 87th Bee Campus USA](#) (*Campus News*)
 - [Southwestern's online presence for Bee Campus USA activities](#)
-

[Green Building Practices at Small Liberal Arts Universities: A Case Study of Southwestern University \(Fall 2019\)](#)

Project Summary: Green building and building certification have mostly positive environmental impacts, but occasionally fail to impose those meaningful changes. Each certification has its benefits and limitations. Pioneering this movement, many universities across the world have signed declarations that require campuses to build their sustainable initiatives. To address these initiatives, universities commonly pursue green building certifications, such as LEED, WELL, and Living Building Challenge. A building does not need to be certified to be sustainable or well-built, but certification ensures a level of

accountability due to the third-party assessment of the project to evaluate its adherence to the standards created. Research and findings are presented and used to create a document of recommendations for improving green building on college campuses around the U.S. A list specifically for Southwestern is provided with outcomes to fit the SU experience.

Completed By: Kathryn Caudell '19, Lois Durant '20, Zane McDurham '20, Dominique Rosario '20, Anthony Trefny '20

[Improving Water Sustainability Through Rainwater Catchment Systems \(Fall 2018\)](#)

Project Summary: Water scarcity has become a serious problem across the world due to increased demand from urbanization, more frequent droughts, and changes in climate patterns. Higher education institutions can implement water conservation efforts due to community-driven initiatives. Students believe that implementing a rainwater catchment irrigation system on campus with the six pre-existing rainwater catchment tanks would achieve water efficiency and conservation goals while allowing Southwestern to become more sustainable in multiple forms. Rainwater irrigation would not only socially, environmentally, and economically benefit the school but could also be used as an educational tool to strengthen water conservation awareness in the community. Our campus's commitment to improving efforts to achieve greater water efficiency, conservation, and management can be utilized as a marketing tool while cost-effectively maintaining the aesthetic appeal of our campus for generations to come.

Completed By: Isabel Calleja '19, Daniela de Souza '19, Hayley Donnelly '19, Taite Drews-Jones '19, Elizabeth Farwell '19, Ryan Moriarty '19, Logan Spalding '19

[Examining the Role of Activism & Action in American College Campuses \(Fall 2018\)](#)

Project Summary: Activism on college campuses has a long history, and within recent years, highly visible student protests surrounding issues such as income inequality, police brutality, indigenous rights, sexual assault, and climate change have gained significant media attention. Although universities often bring together elements that allow student activism to flourish, they also present their own set of internal challenges. With every graduating class, student activist groups contend with changing leadership, reduced membership, and the potential loss of institutional memory. The history of student resistance, conflict, and activism may be erased from, or in some cases co-opted by official institutional records. This creates the challenge of passing knowledge down, along with significant hurdles for any kind of ongoing activist work. This paper responds to this ongoing loss of institutional knowledge by chronicling a counter-history of student and faculty commitment to activism at Southwestern University. Using archival and quantitative research, including interviews with faculty, staff, students, and alumni, interactions between different activist movements on campus, and identifying specific barriers to student activism at SU.

Completed By: Claire Bressette '19, Sean Carroll '19, Hannah Winkler '19, Miranda Wolk '19, Simone Yoxall '19

[Intersectional sustainability and student activism: A framework for achieving social sustainability on university campuses \(Fall 2017\)](#)

Project Summary: In recent decades, universities have made significant progress toward environmental sustainability and have likewise tightened their budgets and economic models in the name of financial sustainability. Although institutions of higher education have failed to address issues of social sustainability and social injustice, which are both increasing in number and severity on college campuses. The article takes a student activist perspective toward university sustainability, with particular reaffirmation of the social sustainability pillar, which can empower students and their allies to raise awareness about the causes of these issues and allow a more constructive environment for collaborative approaches and policy formation on campuses around the world.

Completed By: Keara Hudler '18, Lilly Dennis '18, Nataley Ford '18, Joanna Mendez '18, Muriel DiNella '18

- The research was published in the journal [Education, Citizenship, and Social Justice](#)
 - The published research paper won a [2020 AASHE Campus Sustainability Research Award](#)
-

[A Contemporary History of Georgetown's Switch to Renewable Energy Turning a Red County Green \(Fall 2017\)](#)

Project Summary: With 150-megawatt solar power and 144-megawatt wind power agreements, Georgetown Utility Systems became one of the largest renewable-sourced municipally-owned utilities in the United States. As a city with a well-known reputation

for conservative politics, Georgetown complicates the assumption that support for renewable energy is a progressive environmentalist position. This paper aims to create a contemporary history of Georgetown's switch to renewable energy. Topics in this article include an analysis of the political economy of Texas renewable energy, an explanation of the cultural, economic, and logistical context of Georgetown's decision to switch to renewable energy, and a discussion of whether or not Georgetown may serve as an example to other municipalities weighing the decision to go green.

Completed By: Peri Kincaid '18, Carly Ammel '18, Lucas Evans '18, Zoya Goodwin '18, and Crystal Webster '18

[Southwestern University is Branching Out: Achieving Tree Campus USA Certification \(Spring 2017\)](#)

Project Summary: The Tree Campus USA initiative is a nationwide certification for college campuses through the Arbor Day Foundation. After reviewing the impact of establishing a Tree Campus USA certification on colleges and universities around the nation, the group has worked to add Southwestern University to the Arbor Day Foundation's list of national Tree Campus USA members. Over a hundred colleges and universities have made the promise to take steps forward to improve the health of their campus by maintaining the proper health of their existing trees and creating a "fallow ground" for new ones. For SU, the opportunity to foster stewardship within the campus community, effectively implement the Tree Campus USA standards, and raise awareness for tree health and sustainability will also connect the campus with

the surrounding Georgetown community. To acquire this certification, five standards were completed: creating a campus tree advisory committee; establishing a campus tree-care plan; producing a campus tree program with dedicated annual expenditures; hosting an Arbor Day observance; and completing a service-learning project

Completed By: Rebeca Huteson '17, Alex Morris '17, Colleen Nair '17, Olivia Ruane '17, and Vallery Rusu '17

- [Arbor Day Foundation Honors Southwestern University with 2018 Tree Campus USA® Recognition](#) (*Campus News*)
 - [The Arbor Day Foundation Program Honors Southwestern University with a 2019 Tree Campus USA Award](#) (*Campus News*)
 - [Southwestern University Earns Its Fourth Tree Campus Higher Education® Recognition](#) (*Campus News*)
 - [Southwestern's online presence for Tree Campus Higher Education activities](#)
-

[Sustaining the Southwestern Ecological Laboratory \(Spring 2017\)](#)

Project Summary: The 25-acre Southwestern Ecolab was established in 2014 to provide students in the science and environmental fields with practical, field-based experiences in an interactive environment. Since then, the Ecolab has faced several challenges, including a lack of maintenance and a history of poor management. Additionally, it lacked a clear curricular structure and walking path, decreasing accessibility of the site. These issues combined prevented the formalization of this space and associated classes into the Southwestern curriculum. The group's goals were

to: clean up the Ecolab by removing trash from the site and improving accessibility to the lab itself; identify invasive species and create a species distribution model of one of those species using GIS to create a protocol for future control; develop a curricular structure for Ecolab; secure grant money to continue funding projects; and create a land management and protocol plan to provide guidelines for future research and classroom use.

Completed By: Zach Brown '17, Daniel Buffington '17, Dori Cresanta '17, Thomas Gromatzky '17, Sam Guess '17, Hailey Johnson '17, Austyn Laird '17, Anja Thomsen '17, Farrell Stucky '17, Garrett West '17

[Sustainability and Social Engagement \(Spring 2016\)](#)

Project Summary: The Sustainability and Social Engagement Group at Southwestern University set out to respond to two research questions regarding the university's sustainability efforts: What connects and disconnects SU students to sustainability and social activism? And how can community engagement with sustainable practices and awareness of programs on campus be increased? Students implemented a three-phase plan that began with a review of the scholarly literature on university sustainability and the implementation of a campus-wide sustainability survey. The second phase involved a public relations campaign that combined our findings from the literature with our survey results to educate and inform the campus. The third phase involved a series of presentations to university leaders to raise awareness about student engagement with sustainability issues and the hope of inspiring meaningful changes on campus.

Completed By: Heba Abdel-Rahim '16, Egan Cornachione '16, Patrick Garza '16, Bianca Perez '16, and Dan Ross '16

[Sustainable Infrastructure for the Walzel Gymnasium \(Spring 2016\)](#)

Project Summary: To introduce more sustainable practices to Southwestern University's campus, this project will update the lighting in the Walzel Gymnasium to reduce energy load and increase efficiency. Specifically, this project is a student-organized, student-directed effort to replace Walzel's energy-inefficient incandescent bulbs with more cost-effective and dynamic LED lighting. The direct result of this transition will be a more secure and environmentally-friendly athletic space for Southwestern's students and student-athletes. Even though Walzel gym is only a small portion of the much larger Robertson facility, the proposed lighting changes to this gym are expected to generate enough savings for future capstone groups or the physical plant staff to reinvest those savings into more lighting improvements in Robertson over the next 2-3 years. This will ultimately cut the operational costs of this aging building as well as serve as a springboard for future renovations.

Completed By: Rachel Ehler '16, Dakota McDurham '16, Nick Espino '16, Kron Heilman '16, Dalton Teague '16, and Aaron Moyer '16

[Strengthening the Community Garden \(Spring 2016\)](#)

Project Summary: This project aimed to solidify the Southwestern

University Community Garden as a valuable and sustainable institution of the campus community. Southwestern University Community Garden can be more fully integrated into the larger goals for achieving sustainability to remain a competitive institution of Higher Education, which “encourages contributions toward the well-being of humanity”. The Southwestern University Community Garden is currently an undervalued asset that has historically contributed greatly to that positive change. Founded in 2008, it has been the site of countless projects and studies within the campus community, helped feed the larger Georgetown community through programs like Meals on Wheels, been designated as a wildlife sanctuary, hosted several volunteer workday events, and has contributed to the realization of several of Southwestern University’s core values.

Completed By: Chelsea Allen '16, Gabby Hunt '16, Kali Page '16, Benjamin Galindo '16, Graig Brandley '16, and Lauren Childers '16

[Establishing an SU Office of Sustainability \(2015\)](#)

Project Summary: The research revealed an important pattern among Institutions of Higher Education: colleges and universities all over North America and abroad are prioritizing sustainability. IHEs frequently have significant carbon footprints, and universities should be responsible for reducing that footprint however possible. Implementing this project will allow Southwestern to become a leader among its peers. While Southwestern has enjoyed a great deal of momentum from students, staff, and faculty, the lack of centralized administrative support has led to projects that are often limited and disjointed. Creating an Office of Sustainability would

address this and other challenges to reaching our potential.

Completed By: Cassandra Crabtree '15, Adrienne Dodd '15, Brandee Knight '15, Adriana Guadarrama Lee '15, Travis Kurtz '15, Jonathan Sandoval '15, Keegan Taylor '15, and Rachel Terry '15

[Interactive Memory: Chronicling Sustainability Efforts at Southwestern \(2015\)](#)

Project Summary: Faculty and staff are hired and retire from positions, and students turn over every four years, creating an ever-changing landscape. Each time this happens, valuable knowledge is lost. To recover such information takes trial and error, using valuable time that could be more efficiently spent on further initiatives. To combat the loss of institutional knowledge at Southwestern University, an interactive map was created to chronicle the past and present efforts of students, staff, and faculty. It will serve as a tool for future projects, as well as help alleviate the inefficiencies that occur within sustainability efforts at Southwestern.

Completed By: Brandee Knight '15

- [SUustainability Campus Map](#)
-

[SUustainability Strategic Plan Proposal \(2014\)](#)

Project Summary: Methods or measures for this study will include data collection of current and past metrics of sustainability and environmental awareness, including past carbon inventories, data gathered for the President's Climate Commitment report, the

Tallobres campus-wide sustainability survey, and primary data collection conducted by members of the capstone class. In addition to writing a sustainability strategic plan, the capstone cohort will draft proposals for sustainability initiatives that address current and future needs. An overarching goal of this project is to find the best means by which to promote Southwestern as a symbol of environmental sustainability in Texas.

Completed By: Alec Bergerson '14, Emma Davis '14, Kayla Guentzel '14, Kendall Guidroz '14, Joseph Kyle '14, Brandi Martin '14, Kyle Mathis '14, Thomas Newman '14, Jessica Olson '14, Emily Pawelek '14, Sarah Puffer '14, Ashley Scott '14, Connor Smith '14, Arielle Solcher '14, and Austin Spraez '14

- [Sustainability Strategic Plan \(Results\)](#)
-

[A Sustainable Austin is.. \(2013\)](#)

Project Summary: This study attempts to gain a first glance into how Austinites define sustainability in their city. Students asked Austinites to complete the sentence "A sustainable Austin is..." to determine what is most important to each citizen - information that can be relayed to city leaders to assist in developing policy measures that fit the needs and concerns of Austinites. This study is built on the premise that public policy works best when it is built upon clear feedback from an engaged citizenry. By providing a platform for Austinites to define sustainability for their city, students were able to gain some insight into that diverse vision.

Completed By: Audrey Calhoun '13, Katie Campbell '13, Elise DiNuzzo '13, Elizabeth Funk '13, Liliana Galvez '13, Austin Gentry '13, Cate Jones '13, and Grady Sampley '13

- [A Sustainable Austin Study - Toward a Public Understanding of Sustainability in Austin \(Results\)](#)
-

[The Local Food Landscape in Central Texas: Access, Obstacles, and Consumer Motivations \(2012\)](#)

Project Summary: This capstone examines the specific issues related to local food networks and sustainable agriculture in Central Texas, providing historical information, a review of scholarly literature, and methodologies necessary to justify and carry out a study of local, sustainable food networks in Central Texas. Through the process of background research and development of a literature review of farm-direct markets, students identified key trends and patterns that required further empirical research. This proposal documents that process, suggesting a qualitative mixed-methods approach that examines consumer access and restrictions to local, sustainable food while also investigating several trends of three Central Texas farmers' markets.

Completed By: Katherine Castles '12, Jillian Curran '12, Harrison Glaser '12, Ellen Hoyer '12, Whitney Moore '12, Lisa Morse '12, Molly O'Hara '12, and Ben Parafina '12

- [Farmers' Market Survey & Price Comparison Study \(Results\)](#)